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Sem-II B.Tech 1st Year

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Assignment - 3

Section - A

Q1 Explain the constructor method how it differs from other member functions.

Ans - It is a special member function of class that is automatically called when an object is created. It is used to initialize the object.

Constructor

- Same name as class
- No return type
- Called Automatically

Member functions

- Different name
- Has return type
- Called manually

Q2 Difference b/w default and parameterized constructors.

Default constructor

- Takes no arguments
- Assign default values
- Ex - Student()

Parameterized Constructor

Take arguments

Assigns user-defined value

Ex - Student(int, string)

Q3. Why is constructor used under public access specifier?

Ans - Constructors are declared as public so that object of the class can be created from outside the class. If it is private, obj. creation will not be possible.

Q4. Complete the code (using initializer list)

```
#include <iostream>
using namespace std;
```

```
class Student {
    int roll-number = 101;
    string name = "Pragati";
```

```
public:
    void display () {
        cout << roll-number << " " << name
```

```
    }
```

```
};
```

```
int main() {
    Student s;
    s.display ();
    return 0;
```

```
}
```


Q5 What do you understand by initialization of objects?

Any Object initialization means assigning initial values to the data member of an object at the time of its creation.

Section - B

Q1 Write a Program to depict the function of unary operators overloading.

Ans- Unary operators overloading is used to redefine unary operators like ++, --, - for user-defined class.

```
#include <iostream>
using namespace std;
```

```
class Number {
    int value;
```

```
public:
```

```
    Number (int v) {
        value = v;
```

```
    }
```

```
    void operator --() {
        value = -value;
```

```
    }
```

```
void display () {
    cout << "value:" << value << endl;
}
```

```
};
```

```
int main () {
    Number n(10);
    n.display ();
```

```
-n;
```

```
    n.display ();
    return 0;
}
```

```
}
```

- The operator-() function overloads the unary minus operator

- It changes the sign of the value.

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Q7 Write a program to implement binary operator overloading?

Ans - Binary operators works with two operators like +, -, *, /

```
#include <iostream>
using namespace std;
```

```
class number {
    int value;
```

```
public:
```

```
    number (int v) {
        value = v;
```

```
}
```

```
    number operator+ (number n) {
        return number (value + n.value);
```

```
}
```

```
    void display () {
```

```
        cout << "value:" << value << endl;
```

```
}
```

```
};
```



```
int main () {
```

```
    Number n1 (10), n2 (20);
```

```
    Number n3 = n1 + n2; // calling overloaded +
```

```
    n3.display ();
```

```
    return 0;
```

```
}
```

Explanation:

- operator +() Add two objects
- Return a new object with sum

Q3 Differentiate between a constructor and a destructor.

My Constructor

Destructor

- Initializes object

Destroys object

- called when object is created

Called when object is destroyed

- Same name as class

Same name with ~ symbol

- Can have parameters

Cannot have parameter

- Can be overloaded

Cannot be overloaded

- Ex - Student()

Ex - ~Student()

Part-C

Q1 What is the function overloading? What are the advantages of function overloading? Write a program to calculate the area of circle, area square and area of triangle using function overloading.

Ans-1 Function overloading means defining multiple functions with the same name but different parameters.

Advantages

- Improves code readability
- Reduces function name complexity
- Supports compile time polymorphism
- Reusability of functions name

```
#include <iostream>
using namespace std;
```

```
// Area of circle
float area (float r) {
    return 3.14 * r * r;
}
```

```
// Area of Square
```


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int area(int side) {
 return side * side;
}

// Area of Triangle
 float area(float base, float height) {
 return 0.5 * base * height;
}

int main() {
 cout << "Area of circle : " << area(5.0) << endl;
 cout << "Area of square : " << area(4) << endl;
 cout << "Area of Triangle : " << area(5.0, 2.0) << endl;
 return 0;
}

Q2 What is operator overloading write a program to add two complex numbers using operator overloading.

Ans Operator overloading allows like allow operators like +, -, * to work with user defined data types.

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```
#include <iostream>
using namespace std;
```

```
class Complex {
    int r, i;
public:
```

```
    Complex (int r=0, int i=0) { r=r; i=i; }
```

```
    Complex operator + (Complex c) {
        return Complex (r+c.r, i+c.i);
    }
```

```
}
```

```
void show () {
```

```
    cout << r << " + " << i << "i";
```

```
}
```

```
};
```

```
int main () {
```

```
    Complex c1(2,3), c2(4,5), c3;
```

```
    c3 = c1 + c2;
```

```
    c3.show ();
```

```
}
```

Q3 Design a class student in C++ with the following data members: roll no, name and marks in three subject.

```
#include <iostream>
using namespace std;
```

```
class Student {
    int roll;
    string name;
    float m1, m2, m3;
```

```
public:
```

```
// Default constructor
```

```
Student () {
```

```
    cout << "Enter Roll Number:" ;
```

```
    cin >> roll;
```

```
    cout << "Enter Name:" ;
```

```
    cin >> name;
```

```
    cout << "Enter marks of 3 subjects;" ;
```

```
    cin >> m1 >> m2 >> m3;
```

```
}
```



```

    ... avg;
    Student Details ---
    cout << "Roll:" << roll << endl;
    cout << "Name:" << name << endl;
    cout << "Average:" << average << endl;
    cout << "Percentage:" << percentage << endl;
    cout << "y.o." << endl;
  
```

```

int main() {
  Student s;
  s.display();
  return 0;
}
  
```

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